

Index calculus for ECDLP: what the pipeline is made of and what each piece assumes

Date: 2026-10-06. Status: synthesis (no new runs, no new curves, no new candidates). Base commit: 6350e7f7 (origin/main); every in-repo citation below was verified against that tree. Claims that rest on literature alone are marked recalled. Location: `experiments/ic-assumption-graph/`. Visuals: `ic_pipeline.dot` (overview) and `ic_assumptions.dot` (full graph), rendered to `figures/*.svg` and `figures/*.pdf`. PDF of this report: `REPORT-2026-10-06.pdf` (built from this file; contains both figures).

Claim labels used below: **[proved]** (theorem or per-instance check), **[derived]** (follows from cited proved results), **[measured]** (toy-curve measurement in a frozen receipt), **[heuristic]** (unproved working assumption, sometimes with toy measurements), **[proposed]** (design not yet run), **[open]** (no bound known), **[conditional]** (needs GRH/smoothness or similar), **[refuted]** (over-optimistic as stated; counterexamples or measurements against it). Proposals and extrapolations are never stated as measurements.

1. Question, scope, method

Question. Which algorithms comprise index calculus (IC) for the elliptic-curve discrete logarithm problem (ECDLP), and which assumptions does each algorithm rest on?

Scope. The seven IC stages in `AGENTS.md` (setup, factor base, point decomposition, relation collection, relation linear algebra, target descent, verification) plus optional isogeny transport and the paired Pollard-rho baseline. Both binary-extension curves (Koblitz and subfield families, including ECC2K-130) and prime-field curves (generic, $j = 0$, $j = 1728$, tower bases) are covered. The graph names concrete algorithm families, not one frozen candidate: no `IC1...` ID is issued here (`candidate_id: null`), no factor-base count is inferred, and no speedup is claimed.

Method. Survey of this repository's IC docs, experiments, and code at the base commit, plus the sibling `crypto` study repository's frozen IC notes where this repo points to them, and the classic IC/isogeny literature those files cite. Repo sources below were read in full at 6350e7f7; literature entries are marked recalled where no in-tree full-text extract was read, following the `source_read` convention (`full text / abstract / metadata`; `metadata` means the claim is recalled and flagged, not verified here). Negative and inconclusive findings are kept: Sections 5 and 7 list what failed, what is open, and what was checked.

Curves referenced (no new curves). Examples only, with full immutable IDs from `experiments/ic-candidate-catalog/`

- ECC2K-130 polynomial-basis source: `EC1N131Ckb1h136f03e58c98` (`ecc2k130_pb`), $y^2+xy=x^3+1$ over $\text{GF}(2^{131})$, subgroup order 680564733841876926932320129493409985129, cofactor 4.
- Its degree-263 descendant: `EC1N131Cbinh833014327b07` (`ecc2k130_d263_1`), same field, verified descending route `IW1E263d1hadee4e69fa3d` via edge `e263d1_df508c4fc9a8` (`experiments/ic-candidate-catalog/isog`).
- Toy Koblitz N13: `EC1N13Ckb1h15003cfcd23` (`toy13_kb1`).

Runs. None. This is a synthesis of existing receipts; there are no new workload or run IDs. Measured numbers quoted are cited to the frozen file that holds them, never recomputed here. No Sage jobs were run for this report, so no `./sage --runtime-info` receipt applies; any follow-up Sage work must use the repository `./sage` launcher per `AGENTS.md`.

2. How to read the graph

- **Figure 1 (overview)** shows the pipeline backbone: `SETUP -> FB -> PDP -> RC -> LA -> TD -> VERIFY -> ECDLP solved`, with `ISO` as an optional input to `FB/PDP` and `RHO` as the paired one-target comparator. It is the map; Figure 2 is the territory.
- **Figure 2 (full graph)** puts each stage in one cluster with its algorithms (white boxes) and assumptions (ellipses). Ellipse color is epistemic status: green = proved or checked per instance, yellow = heuristic with toy-curve measurements, orange = open (no bound), red = over-optimistic or refuted as stated, violet = conditional (GRH/smoothness). Solid edges are comprises/requires; dashed edges are optional, shared-heuristic, or comparison links laid out with `constraint=false` so the pipeline order dominates.

Cross-cutting links (smoothness shared by PDP/RC/TD, the orbit fold shared by FB and rho, verification reusing learned-log and transport checks) are dashed for that reason.

- **Files.** `ic_pipeline.dot` and `ic_assumptions.dot` are the editable sources; `figures/ic_pipeline.svg`, `figures/ic_assumptions.svg` are the vector renderings (zoom in the SVG for small text); `figures/*.pdf` are the same figures for LaTeX inclusion. Build with `dot -Tsvg X.dot -o figures/X.svg` and `dot -Tpdf X.dot -o figures/X.pdf` (Graphviz 2.43.0, this report).

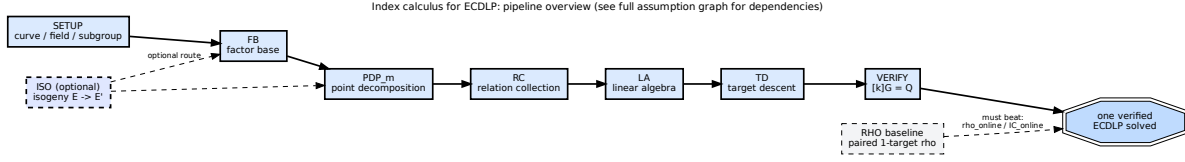


Figure 1: Pipeline overview. Seven stages plus optional isogeny and the paired rho baseline.

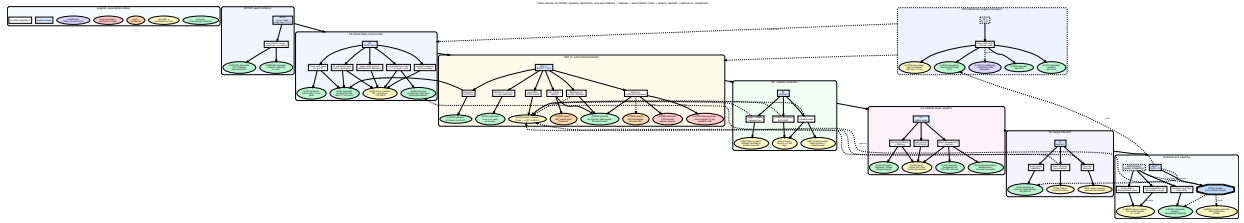


Figure 2: Full algorithm/assumption graph. Ellipse color is status (green proved, yellow heuristic+measured, orange open, red refuted, violet conditional). See the SVG for zoomable text.

3. Stage by stage: algorithms and their assumptions

3.1 SETUP: exact instance

- **Exact field / curve / subgroup / encoding** (white box). Binds the field representation, defining modulus, Weierstrass model, subgroup order r , cofactor, generator G , and element encoding before anything is timed.
 - **A-SETUP1 [proved]**: the EC1...h... curve ID is SHA-256 over the canonical `{"field": ..., "curve": ...}` record (sorted-key compact JSON), excluding the ID itself and any run data (AGENTS.md; curves.yaml identity_rule). Alias and route metadata never change the ID.
 - **A-SETUP2 [proved]**: cofactor clearing maps any curve point into the prime-order subgroup G generates; subgroup membership is then exact (suite/docs/ic/README.md § subfield curves; pdp-degree-heuristics usable-point rule).

3.2 FB: factor-base construction

- **F2-subspace base** (prefix / geometric): $F_V = \{P : x(P) \text{ in } V\}$ for an 1-dimensional subspace V , e.g. $\text{span}\{1, z, \dots, z^{(1-1)}\}$ (experiments/pdp-degree-heuristics/README.md §2).
 - **A-FB1 [proved]**: for prime n , $\dim V^{(2)} \geq 21-1$, and below $n-1$ only geometric progressions $c \cdot \text{span}\{1, g, \dots, g^{(1-1)}\}$ attain it (Hou-Leung- Xiang field Kneser; Bachoc-Serra-Zemor field Vosper; cited and applied in pdp-degree-heuristics §3). Consequence **[derived]**: progressions minimize the product profile and hence the PDP unknown count.
 - **A-FB2 [heuristic + measured]**: a random abscissa lifts with probability about one half and a lifted point lands in $\langle G \rangle$ with probability about one half (cofactor and trace dependent). Receipts record usable B , geometric count, strict-subgroup count, and folded columns separately; B is never inferred from 1 (AGENTS.md; pdp-degree-heuristics §2).

- **Trace-zero base** (geomtrace, kertrace): V inside $\ker \text{Tr}$.
 - **A-FB4 [proved]**: on $y^2+xy=x^3+b$, every point of the odd-order subgroup is a double, so $V \subset \ker \text{Tr}$ puts the base in 2E and doubles the yield at the same degree (except near excess zero where the vacuous trace row costs one degree; `pdp-degree-heuristics` §§4-5,8; Kusters-Yeo [metadata] for the linear-equation mechanism).
- **Automorphism-orbit base** (prime fields): whole Aut orbits ($\{+-1\}$, μ_4 , μ_6) so $w*m$ points carry m unknowns (`suite/docs/ic/README.md` § prime-field curves).
 - **A-FB3 [proved with per-instance checks]**: $\psi(G) = [\lambda]G$ is checked on the subgroup point (sign branch fixed empirically), and orbit folding preserves the relation lattice only with that check plus cofactor handling (`docs/ALGORITHMS.md` § GLV; `suite/docs/ic/README.md`). The older $j=0$ module kept representatives but discarded non-representative images, so its orbits added size without coverage **[measured]** (8.6x vs 8.7x predicted after the fix).
- **Tower base** (prime, Petit-Kusters-Messeng): algebraic factor base over an extension tower (`suite/docs/ic/Framework.md` §8).
 - Coverage is a **heuristic** (A-FB2 family); the solver-axis test that would decide whether to build the oracle end to end is open (pilot FFD flat, round-2 slow rise; no oracle built).
- **Random subgroup base**: pseudo-random abscissae, keep $[h]P$.
 - Rests on A-FB2 only; needs no linear structure, so it suits the pair-table oracle and is refused by Gröbner/SAT oracles that need a subspace (`suite/docs/ic/README.md` § choosing a base).

3.3 PDP_m: point decomposition

Given R , return m base points summing to it (or `None`). Every returned decomposition is re-added in the group before it becomes a relation (`suite/docs/ic/README.md`).

- **Semaev summation polynomials** $S_{\{m+1\}}$ (plus symmetrized / chained-S3 forms; Semaev 2004 [metadata]; Faugère-Huot-Joux-Renault-Vitse 2014 [metadata]; Shantz-Teske 2013 [metadata]).
 - **A-PDP1 [proved]**: $S_{\{m+1\}}(x_1, \dots, x_m, x_R) = 0$ iff the corresponding points sum to R up to sign (Semaev; repo re-verifies by re-addition and by exhaustive cross-checks, e.g. `pdp-degree-heuristics` §9: 73,528/73,528 oracle agreements).
- **Weil descent** to an MQ/Boolean system (binary) or $\mathbb{F}_p$ system (prime).
 - **A-PDP2 [proved with checks]**: descent is sound, but solutions name abscissae over the algebraic closure: lifts can fail (twist points) or hit $x=0$ (2-torsion). Receipts count `lift_failures` and `unliftable_systems` apart from `did not decompose` (`suite/docs/ic/Framework.md` §3; `volcano-ic` § results: 146 spurious solutions rejected at lift time against 28932 expected hits).
 - Yield=Integer models: **A-PDP3 [heuristic + measured]**: $P(\text{decomposable})$ from $|F|^m/(m! \cdot r)$ or the ψ -class count T_{ψ}/r plus a Poisson/birthday step. The naive $|F|^m/\#E$ is off by up to 2.2x; the ψ -class mean is exact for $m=2$ and the Poisson step lands within 2-3% (`pdp-degree-heuristics` §5); `volcano-ic` birthday model within 0.0046 (`experiments/volcano-ic/README.md`).
- **Pair-table / MITM oracle** (meet in the middle over pair sums; $m=2..4$).
 - Rests on A-PDP3 for yield; cost is exact table work ($|F|^{2/2}$ pairs, folded variants) plus lookups. It is the reference row algebraic oracles must beat (`suite/docs/ic/Framework.md` §4).
- **Gröbner family** (Buchberger, F4/F5, Matrix-F4/F5, MutantXL/MXL, XL, Crossbred; Faugère 1999/2002 [metadata]; Joux-Vitse [metadata]).
 - **A-PDP4 [refuted as stated]**: the Petit-Quisquater first-fall-degree assumption that solving degree tracks FFD with $O(\log n)$ growth (Petit-Quisquater 2012 [metadata/abstract]). Counter-evidence: Galbraith- Gebregiyorgis 2014 [metadata], Huang-Kusters-Yeo 2015 [metadata], Kusters-Yeo 2015 [metadata], and repo measurements (FFD 2 for all $m=2$ and 3 for all $m=3$ toy systems while solving degree diverges; `crypto/research/notes/index-calculus/RESEARCH_IC_BOUNDARY_LEDGER.md` §§5,14-17; `pdp-degree-heuristics` §5: FFD is the base degree for 95% of queries whatever the base).
 - **A-PDP5 [refuted as a cost proxy]**: the semi-regular (Bardet-Faugère- Salvy) D_{reg} predicts Gröbner cost. For two-summand Weil systems the homogeneous D_{reg} is exactly 1+1 (bilinear block structure) while D_{solve} is 2-4; the prediction misses by 0.5-1.2 degrees on aver-

- age (pdp-degree-heuristics §§4-5). The structure-only excess rule $D = 2 + [e < -3.5] + [e < -8.5]$ (MAE 0.06, leave-one-n-out 0.08) replaces it for $m=2$ **[measured]**.
- **A-PDP6 [open]**: last-fall-degree bounds cost (Huang-Kiltz-Petit 2015 [metadata]). No repo implementation establishes it as a predictor.
 - **SAT / CDCL / WDSat** (Trimoska-Ionica-Dequen 2019 [metadata]).
 - **A-PDP7 [open]**: descended CNF-XOR scales past Gröbner. Measured on the same systems, CryptoMiniSat gains only ~10% from the product profile where Gröbner gains 10-100x (pdp-degree-heuristics §5). No theoretical bound is claimed.
 - **Linearized half-trace oracle** (PDP2ht; Courtois 2016 [metadata]; Karabina 2015 [metadata]).
 - **A-PDP8 [proved]**: S_3 is F2-linear in (e_1, e_2) , so two-point PDP over V is a linear system in $\dim V + \dim V^{\wedge(2)}$ unknowns plus half-trace splits (Courtois; Karabina; the same S_3 linear structure underlies the linearization-rank analysis in pdp-degree-heuristics §3). Needs $\dim V + \dim V^{\wedge(2)} \leq n$ (A-FB1 gives $31-1 \leq n$ for progressions) **[derived]**. No in-tree implementation or exactness measurement exists at this base commit; the oracle is mapped here from the literature, and any cost claim for it stays **[proposed]** until measured in-tree.

3.4 RC: relation collection

- **Random sampling** ($[a]G + [b]Q$ per trial, $\sim 3 \log_2 r$ adds) and **r-adding walk** (one add per trial; Teske [metadata]).
 - **A-RC1 [heuristic + measured + guard]**: walk targets are treated as uniform, but two rows with the same factor-base part and different (a, b) pin the log by collision, not by relation (24.4% of unguarded two-summand runs end this way; 82% when walk segments share one jump table). The guard (one hash insert per target, priced as a lookup) is mandatory (suite/docs/ic/README.md § boundary ledger round 2).
 - Yield per trial rests on A-PDP3; stopping rests on rank (below).
- **Single / double large-prime variation** (Thomé, Gaudry-Thomé-Diem [metadata]).
 - **A-RC2 [heuristic + measured]**: leftovers meet with birthday probability and the large-prime graph connects. Prime-orbit runs learn $\sim N$ large-prime logs from N differences; descent then hits after $\sim r/(w \cdot L)$ differences (suite/docs/ic/README.md § prime-field curves).
- **Rank accumulation.**
 - **A-RC3 / A-LA4 [heuristic with a proved systematic shortfall]**: $B + \epsilon$ ps relations are expected to give full rank, but on cofactor-4 curves a two-summand relation space has rank $C-1$ unless the base holds a 4-torsion point or lies in $\ker \text{Tr}$ (class-in- $E / \langle G \rangle \cong \mathbb{Z}/4$ kernel; pdp-degree-heuristics §7). Collectors must stop at the achievable rank (computed via `monitor.achievable_rank`), solve modulo the kernel, and verify by fresh descents **[measured: 36 collections, 108/108 descended logs verified]**.

3.5 LA: relation linear algebra

Matrix LA here is the final sparse relation-matrix solve, not a PDP Macaulay reduction (which belongs to PDP per AGENTS.md).

- **Structured Gauss + filtering** (singleton/cliue removal, Markowitz pivot; Odlyzko [metadata]).
 - **A-LA3 [proved]**: eliminations are recorded and back-substituted exactly (suite/docs/ic/README.md § linear algebra).
- **Block Wiedemann / Lanczos** (Coppersmith, Kaltofen-Saunders, LaMacchia- Odlyzko [metadata]).
 - **A-LA1 [proved probabilistic]**: random projections and the Krylov + Berlekamp-Massey step succeed with the standard probabilistic guarantees, conditional on the field/modulus; reports carry core dimension, sequence length, and products.
- **Gain-graph** (1-/2-term relations): exact component-wise solve, every column certified by $[x]G == P$ (suite/docs/ic/README.md § prime).
- **Modulus handling.**
 - **A-LA2 [proved]**: prime r solves directly; $p-1 = \prod q^e$ (classical $\mathbb{F}_p^*$ IC) or composite r goes via Pohlig-Hellman per prime power plus Hensel lift and CRT, with every factor-base log

- verified before use (docs/ALGORITHMS.md § index calculus in $(\mathbb{Z}/p\mathbb{Z})^*$).
- Cost dominance [**measured**]: on $E(F_{p^3})$ the relation phase grows as $n^{1/3}$ but plain LA as $n^{0.68}$, so whole-method S bottoms near $200x$ rho and rises after; the double-large-prime variation grows as $n^{4/9}$ with crossover past 2^{230} (suite/docs/ECDLP_ATTACK_MATRIX.md §10b, citing the crypto residual-walks note).

3.6 TD: target descent and recovery

- **Direct recovery** (one relation over known column logs) and **recursive descent** (Joux-Lercier style [metadata]; GGM small-degree descent [metadata]).
 - **A-TD3 [heuristic]**: rerandomization $Q + [a]G$ (uniform) or additive walk $Q + i[a_0]G$ covers the subgroup; the manifest records which (AGENTS.md TDpdp).
 - **A-TD1 [heuristic + measured]**: recursive subproblems are strictly smaller and terminate. Toy descents verify (e.g. pdp-degree-heuristics §7: 108/108; prime-orbit 64-target batches all recovered), but no general termination bound is claimed.
- **Learning database** (ordered descent adding verified differences).
 - **A-TD2 [proved with checks]**: a learned difference has known log only because $[d]G == Q$ was checked; tampered/mismatched databases are rejected, never trusted (suite/docs/ic/README.md §§ logs/solve, prime).

3.7 ISO (optional): isogeny transport

ISO1 requires an explicit verified map, ordered edge links, subgroup/log transport, and charged route costs; anything less stays a proposal (AGENTS.md). In characteristic two, degree-2 edges carry no ordinary V2 level label (VOLCANO_NAMING.md).

- **Vélu / sqrt-Vélu + volcano walk** (Vélu [metadata]; Kohel thesis [metadata]; Fouquet-Morain [metadata]; Sutherland [metadata]; Bernstein-De Feo-Leroux-Smith sqrt-Vélu [metadata]).
 - **A-ISO1 [proved]**: ordinary components of the 1-isogeny graph are 1-volcanoes; depth is the 1-adic conductor valuation; horizontal/ descending counts follow Kohel/Sutherland [metadata].
 - **A-ISO2 [per-edge, checked]**: kernel rationality must be proved per edge. For ECC2K-130 the registry holds one verified descending edge, `e263d1_df508c4fc9a8`, with map artifact, kernel certificate, and subgroup-transport certificate, forming verified route `IW1E263d1hadee4e69fa3d` from V263L0 to V263L1 (isogeny_routes.json, curves.yaml, and the route manifest in experiments/koblitz-polynomial-w-pair-20260925/). The degree-2 and degree-3 records remain `search_only` with no edges or target curve, so they support no ISO1 candidate (isogeny_routes.json).
 - **A-ISO3 [proved with checks]**: if 1 does not divide r (source subgroup disjoint from kernel, per the transport certificate), $\phi(G)$ retains order r and $\phi(Q)$ is the image of the source target; dual/composition checks when available (MEASUREMENT.md § isogeny gate; isogeny_routes.json certificates).
 - **A-ISO4 [conditional + blocked for ECC2K-130]**: Jao-Miller-Venkatesan GRH expander for horizontal walks [metadata] plus smooth-conductor bridging. ECC2K-130's Frobenius conductor in curves.yaml is $38531015900842053623 = 263 * 146505763881528721$ [**derived arithmetic**]; 263 splits in $\mathbb{Q}(\sqrt{-7})$ (volcano-ic README intro). The primality and inertness of the 57-bit cofactor, which would force any isogeny changing that part to have degree divisible by it, are asserted in the literature summaries but have no in-tree proof at this base commit (open check, §8). Vertical navigation costs at least linear in 1 (Sutherland [metadata]; Robert 2022 notes polynomial-in-log 1 vertical movement as open [metadata]). The 262 conductor-263 descendants reachable by one descending 263-isogeny from E0 (volcano-ic README intro) match Kohel's descending-edge count for a split prime [**derived**].
 - **A-ISO5 [proved]**: Vélu $O(1)$ given the kernel; sqrt-Vélu $O(\sqrt{1})$ in the kernel's field of definition [metadata].

Volcano-ic consequence [**measured**]: at $n=19$ (457-curve class modeling the ECC2K-130 263-level), relation yield depends only on the factor base, not the curve (eligible-pair identity on 457/457; birthday model within

0.0046); E0 needs 0.996x descendants’ attempts (95% CI 0.976-1.017); formal `D_reg` is 11 on all 1,371 sampled systems (1+1 for the dim-10 subspace, cf. the bilinear-block explanation in `pdp-degree-heuristics` §5); 310/310 logs verified (`experiments/volcano-ic/README.md`).

3.8 VERIFY and GOAL

- **VERIFY** checks `[k]G == Q` (and `[x]G == P` per column) before anything is called a result. A row without a verified answer is not a result (`AGENTS.md`; `crypto/AGENTS.md` §2). It reuses A-TD2 (learned logs) and, for ISO1, A-ISO3 (transport), hence the dashed links.
- **GOAL** is one verified target’s online wall time paired with rho on the same point and resource envelope (`online_speedup = rho_online / IC_online`). Multi-target amortizations are secondary and never the headline (`AGENTS.md` measurement rules).

3.9 RHO baseline: what IC must beat

- **Teske r-adding walk + distinguished points** (van Oorschot-Wiener [metadata]; Teske [metadata]).
 - **A-BASE2 [heuristic + measured]**: the walk behaves as random; measured $S \approx 1.25$ (whole-group prime order, one thread; `docs/ALGORITHMS.md`), $S \approx 1.3$ in the study unit. Standard deviation of a single rho run is about half its mean.
- **Negation / automorphism fold** (Wiener-Zuccherato, Duursma-Gaudry-Morain, Gallant-Lambert-Vanstone [metadata]; Bernstein-Lange-Schwabe fruitless-cycle handling [metadata]).
 - **A-BASE1 [proved]**: no generic algorithm beats $\Omega(\sqrt{r/|Aut|})$ (Shoup/Nechaev [metadata]); the fold is $\sqrt{m}$ ($\sqrt{3} \approx 1.73$ for $j=0$ over the negation baseline, $\sqrt{2} \approx 1.41$ for $j=1728$; $\sqrt{2n}$ signed-Frobenius on Koblitz). The prime-orbit pipeline at 28-40 bits spends 1.1-1.4x the folded batch-rho expectation [measured]: it is a generic algorithm with the automorphism oracle, not a subexponential step (`suite/docs/ic/README.md` § prime).
- **Precomputation rho** (Bernstein-Lange 2012/2013 [metadata]).
 - Fair online opponent for any IC online figure that excludes setup ($\sim 1^{(1/3)}$ online after $\sim 1^{(2/3)}$ precomputation); §5.3 derives why $m=2$ subspace IC cannot beat it asymptotically.
- **A-BASE3 [measured + pinned]**: everything is priced in group-addition equivalents per $\sqrt{r}$ (S), with native-to- S conversions pinned in `suite/docs/ic/calibration.json`. Per-run re-measurement drifted ratios by up to 3.7x and repriced identical work by up to 8% (round 4 fix; `suite/docs/ic/README.md` § boundary ledger).

4. Positive findings (what is solid)

1. The pipeline decomposes cleanly into stages with exclusive accounting; the graph’s backbone (`SETUP` through `VERIFY`) matches `AGENTS.md`, `MEASUREMENT.md`, and `FRAMEWORK.md` with no missing stage. [derived]
2. Semaev correctness, Weil soundness (modulo lift checks), product-profile optimality, orbit folding (with checks), trace-zero doubling, linearized `S3`, structured-elimination soundness, Wiedemann/Lanczos probabilistic success, composite-modulus handling, learned-log correctness (with checks), volcano structure, Vélú costs, and the generic lower bound are all [proved] (theorem or per-instance certificate).
3. Smoothness/birthday yield, walk uniformity (with guard), large-prime matching, descent termination, rho random-walk behavior, and calibration stability are [heuristic + measured]: each has toy-curve receipts within a few percent or with explicit intervals, and each has a named guard or pin where the naive form fails.
4. The status split itself is a finding: of 32 assumptions in Figure 2, 16 are proved/checked, 11 are heuristic+measured, 2 are open, 2 are refuted as stated, and 1 is conditional (Section 6). The pipeline’s weight rests on heuristics that have been measured, not on unexamined optimism.

5. Negative and inconclusive findings (what failed or stays open)

1. **FFD assumption [refuted as stated]**. Petit-Quisquater-style claims that FFD bounds solving degree with slow growth do not hold uniformly (Galbraith-Gebregiyorgis, Huang-Kosters-Yeo, Kosters-Yeo;

repo FFD 2/3 with diverging solving degree). Any subexponential claim resting on it alone is not currently believed (suite/docs/ECDLP_ATTACK_MATRIX.md §10).

2. **Semi-regular D_{reg} as a cost proxy [refuted]**. Off by 0.5-1.2 degrees on Weil systems; homogeneous $D_{\text{reg}} = 1+1$ while $D_{\text{solve}} = 2-4$. The excess rule replaces it for $m=2$; $m \geq 3$ degree prediction at scale stays **[open]** (pdp-degree-heuristics §§5,8-9).
3. **$m=2$ subspace IC cannot beat generic rho with precomputation asymptotically [derived]**. A linearized k -point oracle needs $\sum_{i \leq k} \dim V^{(i)} \leq n$, and the product bound (A-FB1, in-tree via pdp-degree-heuristics §3) forces $1 \leq 2n/(k(k+1))$, so the relation phase is at least $2^{(2n/3)}$ for every k (at $n=131$, $2^{87.3}$ plus oracle cost, ~ 26 bits above rho's $2^{60.9}$) with online cost $\sim 2^{(n/3)}$, the Bernstein-Lange exponent [metadata]. Below that needs a non-subspace base or a non-linearized oracle. The in-tree measurement of this bound is not present at this base commit; the derivation here rests on the cited in-tree product bound and the recalled linearization literature.
4. **Prime-field 2-decomposition IC is $O(p^{\{3/2\}})$, worse than rho [measured]**. Test curve $y^2=x^3+x+19 \pmod{271}$: both recover the log; rho ~ 21 ops, IC thousands of trials x 16 relations, exactly as predicted (suite/docs/ECDLP_ATTACK_MATRIX.md § IC).
5. **Gaudry-cubic whole-method never beats rho [measured]**. Relation phase $n^{\{1/3\}}$ but LA $n^{\{0.68\}}$; S bottoms near $200x$ rho and rises after; double-large-prime crossover past 2^{230} (suite/docs/ECDLP_ATTACK_MATRIX.md §10b).
6. **ECC2K-130 isogeny escape is blocked [proved route + derived obstruction]**. The registry holds one verified descending route (degree 263, V263L0 to V263L1); degree-2/3 records are **search_only** with no edges. The in-tree conductor value factors as $263 * 146505763881528721$ with 263 split; if the 57-bit cofactor is (as the literature summaries assert) an inert prime, any route changing that part has degree divisible by it, and vertical 1-isogeny costs at least linear in 1 (open check for the in-tree proof, §8). The 263-level adds a verified route, not an IC1 candidate (no base, PDP, matrix, target, or paired rho there).
7. **Inconclusive / open**: last-fall-degree as a predictor, SAT scaling past Gröbner at scale, tower-oracle $m \geq 4$ (out of memory at $N=16$, only $D \geq 7$), $m \geq 3$ degree rules beyond $N=18$, and any transfer from $m \leq 53$ toys to $m=83$ (required gate per crypto/AGENTS.md §8a) or $m=131$. Extrapolations there stay **[proposed]**, never results.

6. Quantitative companion (summaries, not new measurements)

Assumption-status counts in Figure 2 (32 assumption ellipses, legend excluded; counted from ic_assumptions.dot):

status	color	count	share
proved / checked per instance	green	16	50%
heuristic + measured on toys	yellow	11	34%
open (no bound)	orange	2	6%
over-optimistic / refuted as stated	red	2	6%
conditional (GRH/smoothness)	violet	1	3%

Key measured numbers reused (each cited, none remeasured):

quantity	value	source
rho S (whole group, 1 thread)	~ 1.25 (Z_p^* 32-bit); 0.79-1.7 (curves 24-40-bit, negation map)	docs/ALGORITHMS.md
FFD (toy Semaev-Weil)	2 for all $m=2$, 3 for all $m=3$	RESEARCH_IC_BOUNDARY_LEDGER.md §5
D_{solve} vs D_{reg} ($m=2$)	D_{solve} 2-4 vs homogeneous $D_{\text{reg}}=1+1$; excess rule MAE 0.06	pdp-degree-heuristics §§4-5

quantity	value	source
yield model error	Poisson-from- psi within 2-3%; volcano birthday within 0.0046	pdp-degree-heuristics §5; volcano-ic
unguarded collision ends	24.4% of 2-summand runs (82% with shared jumps)	suite/docs/ic/README.md (round 2)
cofactor-4 rank	C-1 unless 4-torsion or trace-zero; 108/108 descents verify	pdp-degree-heuristics §7
prime-orbit vs folded batch rho	1.1-1.4x (0.72-0.89 as ratio) at 28-40 bits, 64 targets	suite/docs/ic/README.md (prime)
Gaudry-cubic floor	S bottoms ~200x rho; DLP crossover past 2^{230}	ECDLP_ATTACK_MATRIX.md §10b
calibration drift before pinning	ratios to 3.70; identical work repriced to 8%	suite/docs/ic/README.md (round 4)
S_3 linear structure (m=2)	pieces span $w=1+\dim V+\dim V^{\vee}(2)$; refutation prob $\sim 1-2^{-(1-e)}$	pdp-degree-heuristics §3
volcano-ic end-to-end	310/310 logs; E0 0.996x descendants (CI 0.976-1.017)	experiments/volcano-ic/README.md

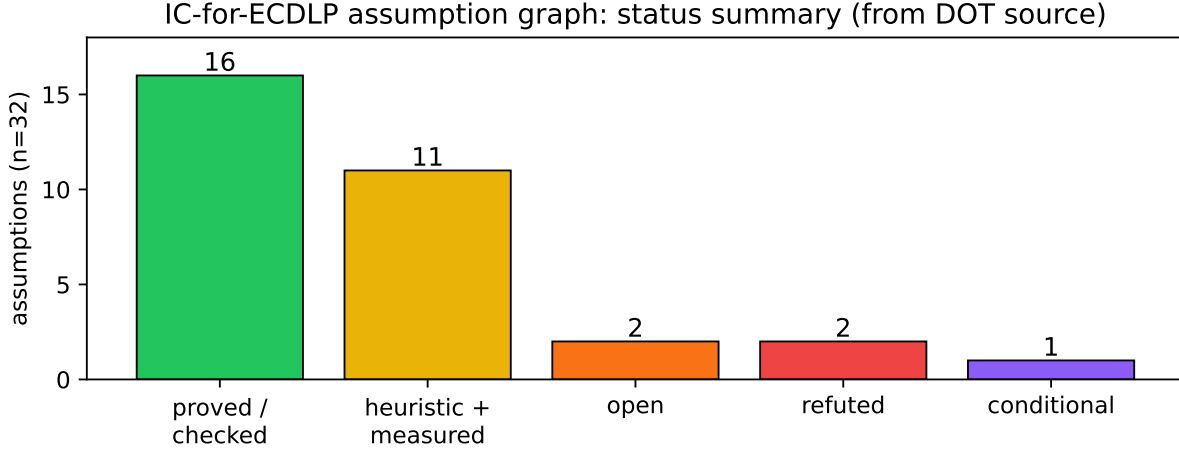
7. Canonical graphs checked (why they remain current)

This change adds a conceptual dependency graph; it asserts no new curve, isogeny edge, factor base, run, or speedup, so no canonical measurement graph moves. Checked:

- **experiments/volcano-ic/figures/census.svg**, **ecdlp.svg** (+ **analyze.py**, **build_report.py** sources): 457-curve census and per-curve ECDLP figures. Unchanged: no new curves, subspaces, or runs; volcano findings reused as cited evidence only.
- **experiments/ic-candidate-catalog/curves.yaml**, **isogeny_routes.json**: exact curve registry and ordered-route registry. Unchanged: no new EC1 nodes or IW1 edges; the 263-route is cited, not extended.
- **experiments/fb-archive/** (**index.csv**, **bases/**, orbit archives): factor-base archive. Unchanged: no new archive rows are added and no digests are cited here, so the **ic-bench** archive gate is unaffected.
- **experiments/pdp-degree-heuristics/** **figures** and **REPORT.md**: PDP-stage profiles (PS1, **candidate_id**: null). Unchanged: profile numbers reused as cited evidence; no new PS1 rows.
- **paper/ecdlp.tex** / built PDF: paper figures and conclusions. Unchanged: no cited conclusion or figure changes, so no rebuild per **paper/README.md**.
- **suite/docs/ECDLP_ATTACK_MATRIX.md** attack taxonomy tree: complementary (attack classes vs IC internals). Unchanged: this graph refines its §10 IC row without contradicting its verdicts.
- Sibling **crypto** scoreboard / leaderboard / lab browser (**docs/index-calculus-scoreboard.html**, **docs/ic-leaderboard.html**, **docs/browser/**): unchanged from here; no new measurements, so §§7-7c obligations do not trigger. Any future measurement PR there must still update every panel per **crypto/AGENTS.md**.

8. Uncertainty and open checks

- Literature entries marked **[metadata]** (Semaev, Diem, Gaudry, Joux-Vitse, Faugère F4/F5, Bardet-Faugère-Salvy, Petit-Quisquater, FPPR, GGM, Huang- Petit, Bernstein-Lange, Shoup/Nechaev, Teske, van Oorschot-Wiener, Vélu, Kohel, Sutherland, Jao-Miller-Venkatesan, Courtois, Karabina, and others) are recalled from the repo's citations and standard knowledge, not re-read here. A reviewer who knows this literature should audit Section 3 against the originals.
- The 57-bit cofactor primality/inertness behind the ECC2K-130 isogeny obstruction (§3.7, §5.6) has no in-tree proof at this base commit; the in-tree facts are the conductor value, its 263 * factorization, the split of 263, and the verified 263-route. Proving or citing the cofactor claim in-tree is an open check.



Green proved/checked; yellow heuristic+measured; orange open; red refuted; violet conditional. Counts from DOT source, not new measurements.

Figure 3: Assumption-status summary. Half the graph’s assumptions are proved or per-instance checked; a third are heuristics with toy measurements; the rest are open, refuted, or conditional. Counts from the DOT source, not new experiments.

- The linearized half-trace oracle (§3.3, A-PDP8) is mapped from the literature with no in-tree implementation at this base commit; wiring it into a measured pipeline (or documenting why not) is an open check.
- The 32-assumption split is a modeling choice: some ellipses bundle several related claims (e.g. A-PDP3 covers both the counting heuristic and the Poisson step; A-RC1 covers uniformity and the guard). Splitting them would raise the count without changing the verdicts.
- Toy-measured heuristics ($n \leq 53$, $N \leq 36$ unknowns) do not transfer to $n=83/131$ without the $m=83$ gate (`crypto/AGENTS.md` §8a) and matched native baseline/candidate reruns. All scale claims above stay predictions.
- No independent red-team review was run for this synthesis. A claim-changing follow-up (new candidate, new route, new speedup) needs the full measurement contract, isolation gate, and review architecture, not just an edit to this graph.

9. Reproduce

```
cd experiments/ic-assumption-graph
dot -Tsvg ic_pipeline.dot -o figures/ic_pipeline.svg
dot -Tpdf ic_pipeline.dot -o figures/ic_pipeline.pdf
dot -Tsvg ic_assumptions.dot -o figures/ic_assumptions.svg
dot -Tpdf ic_assumptions.dot -o figures/ic_assumptions.pdf
python3 figures.py # assumption-status chart -> figures/assumption_status.pdf/.png
pandoc REPORT-2026-10-06.md -o REPORT-2026-10-06.pdf \
  --pdf-engine=pdfelatex -V geometry:margin=1in -V colorlinks:true
```

Graphviz 2.43.0, pandoc 3.1.3, pdfTeX 3.141592653-2.6-1.40.25 (TeX Live 2023/Debian) on this VM. Inspect the SVG in a browser (zoomable) and the PDF for embedded-vector readability before citing.

10. References

Repo sources read in full at the base commit (paths relative to `aburan28/cryptanalysis` unless `crypto`):

- `AGENTS.md`; `experiments/ic-candidate-catalog/README.md`, `MEASUREMENT.md`, `curves.yaml`, `isogeny_routes.json`, `VOLCANO_NAMING.md`; `experiments/koblitz-polynomial-w-pair-20260925/ecc2k130_degre`

- docs/ALGORITHMS.md; suite/docs/ECDLP_ATTACK_MATRIX.md; suite/docs/ic/README.md, FRAMEWORK.md.
- experiments/volcano-ic/README.md; experiments/pdp-degree-heuristics/README.md.
- crypto:research/notes/index-calculus/RESEARCH_IC_BOUNDARY_LEDGER.md (§§5, 10-19, esp. FFD/semi-regular/GAE pinning); crypto:research/notes/index-calculus/RESEARCH_RESIDUAL_WALKS.md (§§9-11, floors and LA dominance, via ECDLP_ATTACK_MATRIX.md §10b); crypto:research/notes/index-calculus/RES crypto:research/notes/index-calculus/RESEARCH_PKM_TOWER_ORACLE.md (§§10-11, via FRAMEWORK.md §8).

Literature (classic IC/isogeny; [metadata] = recalled, flagged, not re-verified here):

- Semaev 2004 (ePrint 2004/031) [metadata]; Diem 2004/2011/2013 [metadata]; Gaudry 2009 [metadata]; Joux-Vitse 2010 [metadata]; Faugère-Perret-Petit- Renault 2012 [metadata]; Petit-Quisquater 2012 [metadata/abstract]; Galbraith-Gebregiyorgis 2014 [metadata]; Huang-Kosters-Yeo 2015 [metadata]; Kosters-Yeo 2015 [metadata]; Petit-Kosters-Messeng 2016 [metadata]; Galbraith-Granger-Merz-Petit 2013 [metadata]; Faugère F4/F5, Bardet-Faugère- Salvy, Courtois-Klimov-Patarin-Shamir XL, Bernstein-Lange precomputation, Shoup/Nechaev generic bounds, Teske, van Oorschot-Wiener, Wiener-Zuccherato, Duursma-Gaudry-Morain, Gallant-Lambert-Vanstone, Bernstein-Lange-Schwabe, Coppersmith/Kaltofen-Saunders Wiedemann, LaMacchia-Odlyzko Lanczos, Vélu, Courtois 2016, Karabina 2015 [all metadata]; Jao-Miller-Venkatesan 2005, Kohel 1996, Sutherland 2013, Galbraith 1999, GHS 2002, Menezes-Teske 2004, Robert 2022, Jao-Soukharev 2010, Bernstein-De Feo-Leroux-Smith 2020, Bisson-Sutherland 2011, Fouquet-Morain 2002 [all metadata].